ITA0608 - MACHINE LEARNING

LAB PROGRAMS

Name: Pravin R

Reg. No: 192421292

Date: 14/07/2026

Experiment 2: Candidate-Elimination Algorithm

Aim

To implement and demonstrate the Candidate-Elimination algorithm to determine the specific and general hypotheses consistent with the given training data.

Procedure

1. Define the training dataset.
2. Initialize the Specific hypothesis (S) and General hypothesis (G).
3. Process each training example.
4. Update S for positive examples and G for negative examples.
5. Display the final Specific and General hypotheses.

Program

```
data = [
['Sunny','Warm','Normal','Strong','Warm','Same','Yes'],
['Sunny','Warm','High','Strong','Warm','Same','Yes'],
['Rainy','Cold','High','Strong','Warm','Change','No'],
['Sunny','Warm','High','Strong','Cool','Change','Yes']
]

c = [x[:-1] for x in data]
t = [x[-1] for x in data]

S = c[0][:]
G = [['?' for _ in S] for _ in S]

for i, h in enumerate(c):
    if t[i] == 'Yes':
        for j in range(len(S)):
            if h[j] != S[j]:
                S[j] = '?'
                G[j][j] = '?'
    else:
        for j in range(len(S)):
            if h[j] != S[j]:
                G[j][j] = S[j]

G = [g for g in G if g != ['?'] * len(S)]

print("Specific Hypothesis:", S)
print("General Hypothesis:")
for g in G:
    print(g)
```

Output

```
Specific Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Result

Thus, the Candidate-Elimination algorithm was implemented successfully, and the specific and general hypotheses consistent with the training data were obtained.